

Upscaler & Vectorizer App – Product and Architecture Design

Executive Summary

This document defines a product and technical design for an image enhancement app that performs two core tasks: (1) neural super-resolution upscaling of raster images, and (2) raster-to-vector conversion to SVG and related formats. The product targets three delivery surfaces that share a common backend: a browser-based web app, a mobile app, and an AI chat bot interface similar to the Topaz Labs Poe bot.^{[1][2][3][4][5][6][7]}

The design reuses the Caliper raster-to-vector engine as the primary vectorization backend and adds a dedicated super-resolution service powered by ONNX Runtime and GPU/CPU acceleration. The app differentiates itself from existing tools by combining high-quality upscaling and deep-learning-assisted vectorization in a single pipeline, exposing both low-friction presets and expert controls, and offering agent-friendly APIs and a chat surface from day one.^{[2][8][9][4][6][7][^1]}

Competitive Landscape

Vectorizer.io (web)

Vectorizer.io is a simple online raster-to-vector converter that accepts JPEG and PNG uploads and outputs scalable vector art such as SVG. Its documented approach is classical: convert pixel color regions into geometric primitives after edge detection across areas of similar brightness or color. The product is optimized for single-image, browser-only workflows with minimal controls and no exposed API for new users.^[^6]

Key lessons:

- A single-screen upload → preview → download flow dramatically reduces friction for casual users.^[^6]
- Classical pipelines can produce acceptable logo-style results but struggle with complex photos, gradients, and noise.

Vectorizer.ai (web)

Vectorizer.ai provides a more advanced cloud raster-to-vector service using a proprietary “Deep Vector Engine” that combines deep learning networks with classical algorithms. It supports full-color tracing, automatically converts bitmaps to SVG/PDF/EPS/DXF, and handles logos, sketches, and photos. Its proprietary “Vector Graph” data structure enables automated edits and localized optimizations beyond conventional vector representations, including rich curve types beyond plain cubic Béziers.^{[9][2]}

The service is fully automatic, requires no user input for basic results, and offers options such as parameterized shape detection (circles, rectangles, stars) and grouping by color or parent shape for easier post-editing. It runs as a closed SaaS with subscription and credit-based plans.^{[10][2]}

Key lessons:

- Deep-learning-assisted tracing significantly improves curve quality, node count, and semantic awareness of shapes compared to purely classical tracers.^{[8][2]}
- Automatic detection of parameterized shapes and structural grouping is a major UX win for designers editing results in downstream tools.^{[2][10]}

Vector Magic and related tools

Vector Magic is a well-known auto-tracer that analyzes images to auto-detect tracing settings and produce full-color vector output, mainly targeting logo and artwork conversion use cases. It focuses on fully automatic tracing with meaningful user-tweakable parameters and offers both online and desktop apps. Other tools like Vectr and Canva lean more toward a full vector editor experience with auxiliary JPG→SVG conversion rather than pure tracing engines.^{[11][12][13]}

Key lessons:

- Auto-detection of appropriate tracing parameters plus a small set of meaningful user controls covers most common workflows.^[12]
- Many users primarily need “fix my client’s low-res PNG logo” and “convert this artwork to scalable vector” workflows; the product should optimize for these.

Topaz Labs (desktop and web)

Topaz Labs specializes in AI-based image enhancement and upscaling, with products like Topaz Gigapixel AI and Topaz Photo AI. Their models upscale images up to 6× (desktop) and up to 8× online, with per-image Autopilot analysis to select model variants (Standard, High Fidelity, Low Resolution, Graphics, Face Recovery, etc.). The tools focus on high-quality detail reconstruction, noise reduction, artifact removal, and specialized handling of AI-generated images.^{[14][15][4][16][7]}

Key lessons:

- Multiple specialized models (photos, low-res scans, graphics, faces) plus an automatic “best model” selection flow is a proven UX pattern.^{[7][14]}
- Constraints like maximum output size (e.g., 32,000 × 32,000 pixels, 1 gigapixel) need to be clearly communicated and enforced.^[16]

Topaz Labs Poe bot (chat surface)

The Topaz Labs Poe bot exposes Topaz upscaling via a chat interface, enabling users to upload an image and apply upscaling with simple commands or a follow-up button from generated images. Users can specify scale factors from 2× to 16×, explicit output dimensions, and whether the source is AI-generated via flags like `--upscale`, `--output_height`, `--output_width`, and `--generated`.^{[3][5]}

Key lessons:

- A text-first interface with clear command flags is effective for power users and integrates well into broader AI workflows.^{[5][3]}
- Tight integration into a host platform (Poe) via one-click “Upscale with Topaz” affordances drives adoption.

Mobile logo maker apps

Popular Android logo maker apps such as “Logo Maker - Logo Creator” focus on template-driven logo creation, layering text and graphical elements, and exporting designs for social media and branding. They offer large sticker and background libraries, multiple fonts, layering, cropping, and basic photo editing tools but typically do not expose raw vectorization pipelines to the user.^{[17][18][^19]}

Key lessons:

- On mobile, users expect immediate visual feedback, drag-and-drop editing, and export to transparent PNG and vector formats.^{[20][17]}
- Vectorization is often implicit (e.g., exporting vector versions of template-based logos) rather than a primary, user-visible feature.

Product Vision and Positioning

Vision

Provide a unified, developer- and designer-friendly platform that accepts any raster image (photo, logo, scan, AI art) and returns a high-quality upscaled raster, a clean SVG vector, or both, with consistent behavior across web, mobile, and chat surfaces.^{[1][8][^2]}

The product aims to be the “Caliper-powered Topaz + Vectorizer in one box”: Caliper serves as the vectorization engine, while a super-resolution pipeline provides state-of-the-art upscaling, and the entire system exposes structured, scriptable APIs suitable for agentic workflows.^{[8][7][^1]}

Target users

- Brand and logo designers needing to convert client-supplied PNG/JPEG logos into clean vector art and high-res outputs.
- Illustrators and digital artists converting sketches and AI-generated art into scalable assets for print or animation.^{[15][8]}
- Developers and AI-tool builders wanting a reliable, scriptable backend for upscaling and vectorization.
- Print shops, merch vendors, and sign makers needing consistent, high-resolution inputs from low-quality uploads.^{[4][12]}

Key differentiators

- **Unified pipeline:** Coherent UX for both upscaling and vectorization, including combined workflows (upscale then vectorize, or vectorize then render at arbitrary resolution).
- **FOSS vector core:** Caliper provides a transparent, auditable, and extensible vectorization engine with a well-documented pipeline and high-performance CPU/GPU/NPU backends.^[^1]
- **Multi-surface by design:** Same backend powers a responsive web app, native-like mobile apps, and a chat bot, with shared presets and behaviors.^{[3][5]}
- **Agent-first APIs:** Structured JSON responses, stable REST/gRPC APIs, and a dedicated chat-bot surface modeled on Poe's Topaz integration.

User Workflows

Core scenarios

1. Logo clean-up and vectorization

- User uploads a low-resolution PNG/JPEG logo.
- App detects image class = logo/flat art via heuristics and optional classifier.
- Pipeline performs mild denoising, color quantization, contour tracing, and path fitting via Caliper.
- Output: SVG (and optional PDF/EPS/DXF), previewed inline with overlays for shape grouping and node count.

2. Photo/scan upscaling

- User uploads a low-resolution photo or scan.
- App selects a photo-optimized super-resolution model and optional denoise/deblur settings.
- User chooses scale factor (2×–6×) or target dimensions, with constraints shown (max dimension, max megapixels).
- Output: high-resolution PNG or TIFF.

3. AI art enhancement

- User imports AI-generated images from other tools, optionally tagging them as AI-generated.
- App selects specialized models tuned for AI-art textures and reduces typical AI artifacts while upscaling.^{[15][7]}
- Output: large, clean images for print or further composition.

4. Sketch to vector

- User uploads a scan or photo of hand-drawn line art.
- Pipeline applies contrast boosting, thresholding, edge detection, and vectorization with strong simplification and path smoothing.^[^8]
- Output: SVG suitable for further editing in vector design tools.

5. Batch processing

- Power users upload a ZIP or directory (web) or select multiple files (mobile).
- App queues jobs and processes them asynchronously, providing per-file results and a downloadable bundle.

Surface-specific UX

Web app

- Single-page app with three primary modes: Upscale, Vectorize, and Hybrid (do both in a defined order).
- Drag-and-drop upload, file picker, and optional pasted image support.
- Live previews with side-by-side comparisons (before/after), zoom, and overlays (edges, paths, color regions).
- Preset panel (Logo, Photo, AI Art, Sketch, Pixel Art) plus advanced sliders and toggles for expert control.
- Job history panel with thumbnails, metadata (scale, model, vectorization options), and re-run/variant actions.

Mobile app

- Similar modes as the web app but optimized for touch: large tap targets, full-screen preview with pinch-zoom.
- Integration with camera roll and camera capture.
- Local caching of recent jobs and offline queueing where possible (for on-device models).
- Export to PNG (including transparent backgrounds), SVG/PDF, and direct share to other apps.

AI bot

- Chat interface accessible via platforms like Poe, Discord, or a custom web chat.
- File upload followed by text commands, e.g.:
 - `/upscale 4x mode=photo`
 - `/vectorize style=logo colors=16`
 - `/hybrid upscale=2x vectorize=logo`
- Bot responds with processed image attachments and concise JSON metadata (scale factor, model, node count, file URLs).
- Short, discoverable command syntax modeled after Topaz's Poe bot flags, including explicit scale, output dimensions, and AI-generated hints.^{[5][3]}

Functional Requirements

Input and output formats

- **Raster input:** PNG, JPEG/JPG, WebP, BMP, TIFF, AVIF, QOI (where supported by underlying libraries).^[^1]
- **Vector output:** SVG (primary), PDF, EPS, DXF for downstream CAD/workflow integration.^{[2][1]}
- **Raster output:** PNG, TIFF (16-bit where applicable), JPEG for web-oriented exports.^[^4]

Upscaling service

- Support scale factors from 2× to at least 6×, with the option to specify a target longest side (subject to a max dimension, e.g., 16,384–32,000 pixels).^{[16][15]}
- Model presets:
 - **Photo:** natural images, portraits.
 - **Low-Res:** heavily compressed or very small source images.
 - **Graphics:** flat colors, UI elements, pixel art.
 - **AI Art:** generative art artifacts, textures.^{[14][7]}
- Optional steps: denoise, deblur, artifact removal, face-aware enhancement (for portraits), and color-space-aware processing.
- Asynchronous processing with job IDs and status polling (REST) or streaming updates (WebSocket/SSE).

Vectorization service (Caliper)

- Integrate Caliper as the primary vectorization backend via a stable C-ABI or Rust crate interface.^[^1]
- Expose profiles aligned with Caliper's Assayer image classes (logo, technical drawing, photo, pixel art) and quantization options.^[^1]

- Allow configuration of:
 - Color quantization method (OKLab k-means, median-cut, octree) and palette size.^[^1]
 - Edge detector (classical Sobel/Scharr vs ML-based HED/BDCN routes).^{[8][1]}
 - Contour simplification strength and corner sensitivity.
 - Primitive fitting (on/off) for circles, rectangles, stars, etc.^{[10][1]}
- Provide a summary of vector output metrics: number of paths, nodes, layers, and estimated file size.

Hybrid workflows

- **Upscale then vectorize:** recommended for tiny logos or icons where vectorization benefits from clearer edges.
- **Vectorize then rasterize at arbitrary resolution:** recommended when the goal is a vector master plus multiple raster sizes; output pipeline renders SVG to requested resolutions.
- UI should present these workflows as explicit options with simple presets.

Batch and API features

- REST API endpoints for upscaling and vectorization operations with JSON metadata and signed URLs for uploads/downloads.
- Batch endpoints that accept multiple files and optional per-file overrides.
- Authentication via API keys or OAuth for third-party integration.
- Rate limiting and tiered quotas per plan.

Non-Functional Requirements

Quality and metrics

- Track quantitative metrics where feasible: PSNR/SSIM for upscaled images versus ground truth in benchmarking suites, and geometric deviation and node counts for vectorization benchmarks.^[^8]
- Provide internal QA tools for side-by-side comparison against baseline algorithms (e.g., classical interpolation, Potrace-style tracing).^{[12][1]}

Performance and scalability

- Latency targets: interactive single-image jobs should complete within a few seconds for typical 1–4 megapixel inputs on GPU-equipped servers.^{[4][8]}
- Scale horizontally by queueing jobs and running multiple worker instances with GPU access.
- For CPU-only deployments, rely on Caliper’s SIMD and Rayon-based concurrency and efficient ONNX CPU backends.^{[8][1]}

Privacy and security

- All uploads are processed over HTTPS and stored with short-lived retention windows unless the user explicitly saves to a library.
- Optional “no storage” mode where images are kept only in memory until processing completes and are not written to persistent storage.
- Clear privacy policy, including handling of user data for any model fine-tuning or future improvements (default to opt-out of training on user assets).

System Architecture

High-level overview

The system is organized into the following major components:

- **API Gateway & Auth:** Fronts all external traffic (web, mobile, bot, third-party API), handles auth, rate limiting, and request validation.
- **Job Orchestrator:** Accepts processing requests, enqueues jobs, and tracks their lifecycle.
- **Upscale Service:** Runs super-resolution models via ONNX Runtime with GPU/CPU/NPU backends.^[^7]
- **Vectorize Service (Caliper):** Wraps the Caliper library and exposes vectorization operations over RPC.^[^1]
- **Storage:** Object storage (e.g., S3-compatible) for original uploads and generated assets; metadata DB (SQL) for job records and user data.
- **Frontend apps:** Web SPA, mobile apps, and chat-bot bridge.

Backend services

Upscale Service

- Implemented in a systems language with strong concurrency and ONNX Runtime binding (Rust or Go with FFI); Rust aligns well with Caliper.
- Hosts one or more super-resolution models:
 - General SR (ESRGAN-like or SwinIR-like architectures).
 - Photo-optimized variant.
 - Graphics/pixel-art variant (preserves sharp edges and flat colors).
 - Optional AI-art-tuned model.
- Uses ONNX Runtime execution providers (CUDA, TensorRT, DirectML, OpenVINO, CoreML, etc.) similar to Caliper’s NPU backend philosophy.^{[7][1]}
- Performs pre-/post-processing: color-space conversion, tiling for large images, and seam-aware tile stitching.

Vectorize Service (Caliper)

- Wraps the Caliper pipeline orchestrator (`caliper-crucible`) and Assayer to select vectorization configurations based on input characteristics and user presets.^[^1]
- Exposes an RPC or gRPC interface like `Vectorize(request) -> VectorizeResponse` with fields for:
 - Input image URL or binary.
 - Desired output formats.
 - Profile (logo/photo/sketch/pixel art).
 - Override options for quantization, edge detection, simplification, primitive fitting.
- Returns URLs to generated vector files plus metrics and optional debug overlays (e.g., edge maps, contour plots).

Job Orchestrator

- Coordinates multi-stage workflows (e.g., upscale → vectorize → rasterize vector) by chaining service calls.
- Maintains job state machine: QUEUED → RUNNING → SUCCEEDED/FAILED.
- Publishes status updates via WebSockets/SSE to the frontend and via progress messages to bot channels.

Data model

Core entities:

- User: identity, plan, quotas.
- Job: type (upscale, vectorize, hybrid), status, timestamps, configuration, and references to assets.
- Asset: original raster, intermediate outputs, final outputs (raster/vector), storage location, format, and metadata.

Frontend and UX Architecture

Web frontend

- Implemented as a SPA using a modern framework (React, SvelteKit, or similar) with:
 - Upload component with drag-drop and clipboard paste.
 - Preview canvases using WebGL/Canvas2D for fast zoom and pan.
 - Form components for presets and advanced options.
 - Job history and status indicators tied to the orchestrator.
- Optional in-browser preview processing for quick feedback (e.g., small crops) using WebAssembly-compiled kernels.

Mobile frontend

- Built using a cross-platform framework (Flutter, React Native, or Kotlin Multiplatform) or native iOS/Android, depending on priorities.
- Uses the same REST/gRPC APIs as the web frontend.
- May optionally ship with on-device SR and vectorization kernels for offline or privacy-sensitive workflows, subject to model size and hardware constraints.

AI bot integration

- Bot bridge service listens for events from the host platform (Poe, Discord, custom chat), fetches attached images, and turns commands into backend job requests.
- Command parser supports flags similar to Topaz's Poe bot: `--upscale`, `--output_width`, `--output_height`, `--generated`, plus vectorization-specific flags (`--vectorize`, `--colors`, `--profile`).^{[3][5]}
- Responses include a short textual summary, one or more image/vector attachments, and a machine-readable metadata blob for downstream agents.

Implementation Phases

Phase 1 – Core backend and web MVP

- Implement Upscale Service with at least one general SR model and a graphics-optimized variant.
- Integrate Caliper as a standalone Vectorize Service with logo- and sketch-optimized presets.^[^1]
- Build API Gateway, Storage, and Job Orchestrator.
- Deliver a basic web UI for single-image upscaling and vectorization with presets and side-by-side previews.

Phase 2 – Quality improvements and hybrid flows

- Add more specialized SR models (photo, AI art) and automatic model selection heuristics.
- Expose advanced vectorization controls (color palette size, primitive fitting, path simplification levels).
- Implement hybrid pipelines (upscale → vectorize, vectorize → rasterize at arbitrary sizes).
- Add batch processing and an initial public REST API.

Phase 3 – Mobile apps and AI bot

- Deliver mobile apps using shared APIs, focused on logo clean-up, quick upscaling, and social media export.
- Implement AI bot integration on at least one chat platform, with documented commands and structured responses.^{[5][3]}

- Add billing, quotas, and usage analytics.

Phase 4 – Optimization and ecosystem

- Optimize GPU and NPU usage; add additional ONNX Runtime execution providers as needed.^{[7][1]}
- Publish SDKs and client libraries (TypeScript, Python, Rust) for developers.
- Consider on-prem or self-hosted editions targeting agencies and enterprises.

Future Extensions

- **Interactive vector editor:** minimal browser-based editor for quick tweaks to Caliper output (grouping, node simplification, shape replacement).
- **Model fine-tuning:** optional opt-in usage of anonymized assets to refine SR and vectorization models for specific domains.
- **Text-to-logo and generative features:** integrate text-to-image/vector models to generate logos and then immediately refine via the same pipelines.^{[21][22][^11]}
- **Third-party integrations:** plugins for Figma, Adobe Illustrator, Affinity Designer, and CMSes that call the backend APIs transparently.

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